

SDS 5380 Social Algorithm Assignment 2
Name: Tracy Chen, Hayley Shin (Group No 7)

Part I: Feed Analysis [ALL CODEs ARE SAVED IN Part1_Analysis.ipynb]

I.1 Data Collection:

By the python coding scripts, the feed for each senator is created, with three different JSON files:

- PartI1_senator_follows.json: stores the social graph of follows for each senator. *Key*: A senator's handle *Value*: A list of dictionaries, where each dictionary contains detailed information about an account that the senator follows.
- PartI1_feeds_data.json: contains the recent activity of every unique account in the follow list. *Key*: handle of a unique account *Value*: A list of post objects posted by that account within the last 24 hours.
- PartI1_posts_data.json: represents the reconstructed user experience. *Key*: A senator's handle *Value*: A single, combined list of posts from all the accounts that senator follows, sorted in reverse-chronological order (newest first).

I.2 "Senators You May Know" Recommendations

Based on the description, we used the triangle-counting approach to recommend three senators (top-3) for each senator if they have not fully followed everyone in the dataset. The output table is attached below:

There's no senators who follow all other senators.

Senators who follow zero other senators in the dataset:

Jeff Merkley (@jeff-merkley.bsky.social); Chris Murphy (@chrismurphyct.bsky.social)
Mark Kelly (@captmarkkelly.bsky.social); John Fetterman (@fetterman.senate.gov)
Maggie Hassan (@hassan.senate.gov); Bernie Sanders (@sanders.senate.gov)
Kirsten Gillibrand (@kirstengillibrand.bsky.social)

There's no senator who is followed by all senators.

Senators followed by zero other senators:

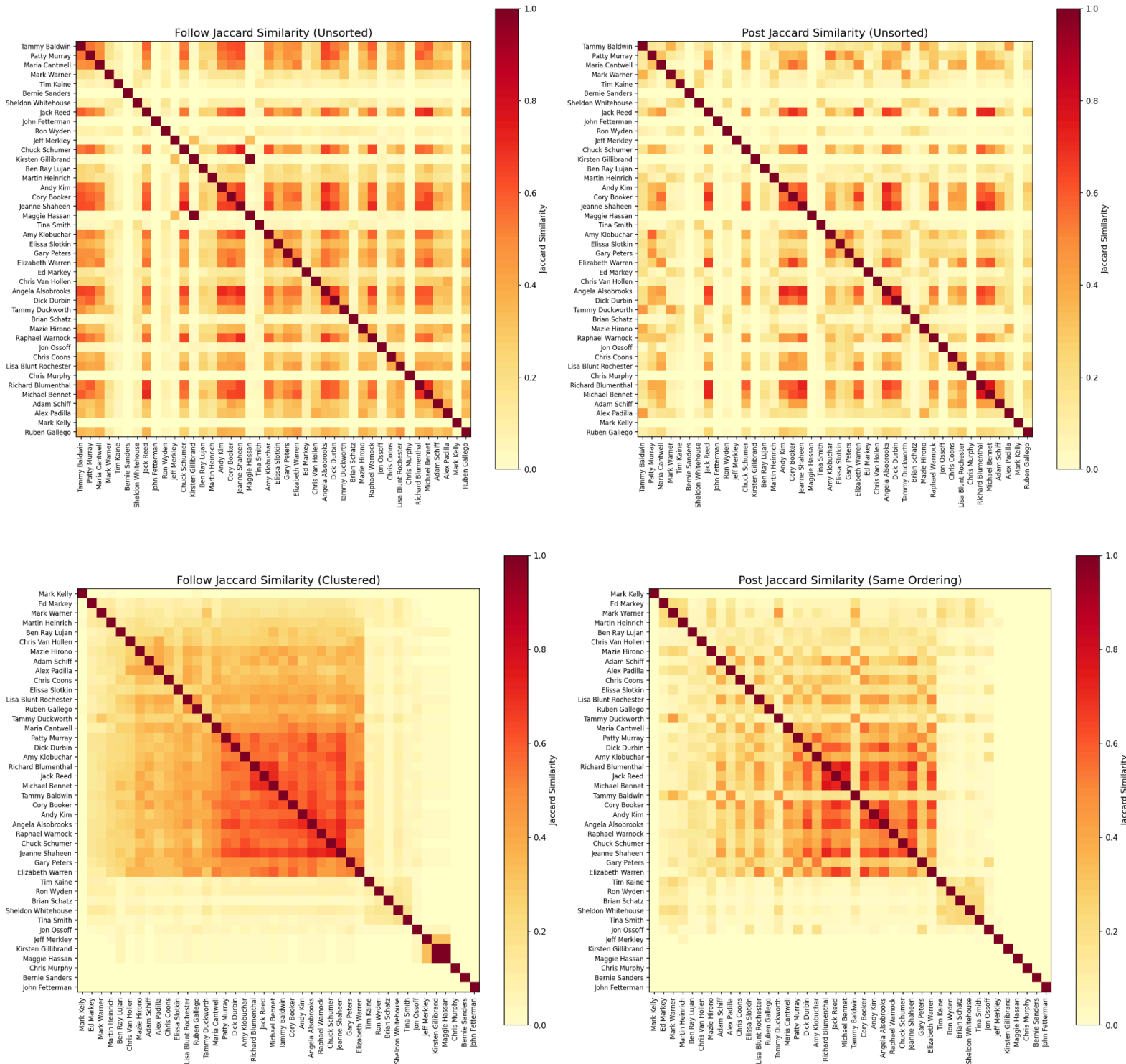
Kirsten Gillibrand (@kirstengillibrand.bsky.social)

Copied table from PartI2_recommendation.csv

Senator	Recommendation 1	Recommendation 2	Recommendation 3
Adam Schiff	Ed Markey (17)	Brian Schatz (17)	Michael Bennet (16)
Alex Padilla	Tammy Baldwin (13)	Adam Schiff (12)	Jeanne Shaheen (12)
Amy Klobuchar	Andy Kim (16)	Elissa Slotkin (16)	Maggie Hassan (4)
Andy Kim	Jeanne Shaheen (19)	Raphael Warnock (15)	Dick Durbin (11)
Angela Alsobrooks	Raphael Warnock (16)	Dick Durbin (12)	Jack Reed (11)
Ben Ray Lujan	Michael Bennet (11)	Cory Booker (11)	Maria Cantwell (11)
Brian Schatz	Tim Kaine (11)	Chris Van Hollen (11)	Elizabeth Warren (11)
Chris Coons	John Fetterman (18)	Tammy Baldwin (16)	Tammy Duckworth (15)
Chris Van Hollen	Tim Kaine (14)	Elizabeth Warren (14)	Tammy Duckworth (13)
Chuck Schumer	Jeff Merkley (5)	Jon Ossoff (1)	
Cory Booker	Raphael Warnock (16)	Elissa Slotkin (13)	Dick Durbin (12)
Dick Durbin	Andy Kim (15)	Jack Reed (11)	Chris Coons (10)
Ed Markey	Andy Kim (15)	Dick Durbin (13)	Jack Reed (11)
Elissa Slotkin	Tammy Duckworth (17)	Amy Klobuchar (17)	Dick Durbin (11)
Elizabeth Warren	Amy Klobuchar (15)	Bernie Sanders (15)	Tammy Duckworth (14)
Gary Peters	Amy Klobuchar (15)	Bernie Sanders (15)	Tammy Duckworth (14)
Jack Reed	Andy Kim (15)	Elissa Slotkin (14)	Dick Durbin (13)
Jeanne Shaheen	Angela Alsobrooks (7)	Jeff Merkley (5)	Mark Kelly (2)
Jon Ossoff	Michael Bennet (1)	Cory Booker (1)	John Fetterman (1)
Lisa Blunt Rochester	John Fetterman (18)	Tammy Baldwin (16)	Adam Schiff (15)
Maria Cantwell	Chuck Schumer (22)	Chris Murphy (16)	Tammy Duckworth (15)
Mark Warner	Cory Booker (18)	Martin Heinrich (18)	Patty Murray (18)
Martin Heinrich	Mark Warner (25)	Bernie Sanders (19)	Lisa Blunt Rochester (17)
Mazie Hirono	Tammy Baldwin (16)	Adam Schiff (15)	Jeanne Shaheen (15)
Michael Bennet	Bernie Sanders (18)	Andy Kim (14)	Elissa Slotkin (14)

Senator	Recommendation 1	Recommendation 2	Recommendation 3
Patty Murray	Dick Durbin (13)	Mazie Hirono (11)	Jack Reed (11)
Raphael Warnock	Chris Coons (11)	Jeff Merkley (5)	Mark Kelly (2)
Richard Blumenthal	Andy Kim (15)	Elissa Slotkin (14)	Jack Reed (12)
Ron Wyden	Elissa Slotkin (15)	Chris Coons (11)	Mazie Hirono (11)
Ruben Gallego	Tammy Baldwin (14)	Adam Schiff (13)	Jeanne Shaheen (13)
Sheldon Whitehouse	Elissa Slotkin (17)	Jeff Merkley (5)	Maggie Hassan (4)
Tammy Baldwin	John Fetterman (27)	Dick Durbin (13)	Jack Reed (12)
Tammy Duckworth	Mazie Hirono (11)	Jack Reed (11)	Chris Coons (10)
Tim Kaine	Mazie Hirono (13)	Jeff Merkley (5)	Maggie Hassan (4)
Tina Smith	Lisa Blunt Rochester (19)	Dick Durbin (13)	Angela Alsobrooks (7)

I.3 Echo Chamber Analysis



Compare and contrast the two matrices:

What do they show? The Follow Jaccard matrix captures the information about which accounts each senator has chosen to follow. It shows a heavily overlapping situation between the followers by senators. The Post Jaccard shows what senators actually saw in a 24 hour window.

Clustering Patterns: Based on the Follow Jaccard graph, there's a core group clustered on the upper-left corner with about 30 senators showing a moderate-to-high floor overlap. And the bottom rows/columns show senators with near-zero similarity to everyone.

Difference between the two matrices: While the cluster structure in the Post Jaccard matrix is still visible, it's lower than what discovered in the Follow Jaccard matrix, shows that shared follows do not guarantee a shared information experience. When the same clustering order is applied to the post matrix, the large warm block becomes more fragmented. This means that even among senators who follow identical accounts, the actual posts viewed can differ substantially. There's a limitation for the Jaccard method we use as discovered by looking at Gillibrand & Hassan: Both senators follow exactly one account. Since their follow sets are identical, their Follow Jaccard is a perfect 1.0. However, this is an artifact of small set size, but neither senator had any posts in their feed during the 24-hour window. This highlights a known limitation of the Jaccard index: it can produce artificially high scores when the underlying sets are very small.

I.4 Reflection Questions

1. A core cluster of ~30 senators shares a substantially overlapping follow graph. However, post-level similarity is consistently lower and more fragmented. Seven senators are clear outliers: Sanders, Fetterman, Murphy, Kelly, Gillibrand, Hassan, and Merkley. They follow very few or zero accounts and are almost entirely disconnected from their colleagues' shared information environment.
2. Feed volume varies enormously. Tina Smith followed 891 accounts and saw 5,212 posts in 24 hours; Jeff Merkley followed 3 and saw 29; six senators had completely empty feeds. This creates a tension between overload and scarcity. Smith faces a firehose impossible to fully read, while someone could realistically consume their entire feed. The variation also affects constituent awareness. Senators with few followers are unlikely to encounter constituent voices or local news, for them, Bluesky might be a broadcasting tool, not a listening tool. Those with larger follow lists encounter more perspectives, though their feeds may still be dominated by a few prolific posters.
3. The most frequently recommended senators are Dick Durbin (in 9 senators' top-3), Elissa Slotkin and Jack Reed (7 each), Andy Kim and Tammy Duckworth (6 each). Durbin's top position indicates he is particularly central in the caucus follow graph. By contrast, the most followed senator Sheldon Whitehouse who is followed by 33 senators rarely appears as recommendations because almost everyone already follows them. This highlights the difference between being a well-connected core node and being a structural bridge. Seven senators follow zero other senators and cannot receive recommendations, since the triangle-counting algorithm requires at least one existing edge. This one-directional relationship suggests they are recognized as important voices by colleagues, even without engaging in the caucus follow network themselves.

Part II: Reply Homophily [ALL CODEs ARE SAVED IN Part2_ReplyHomophilyAnalysis.ipynb]

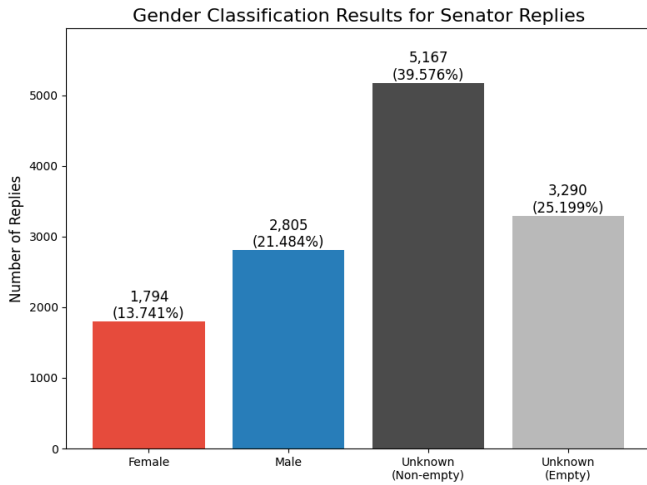
II.1 Data Collection

By the python coding scripts, reply data for each senator is collected and stored in JSON files:

- PartII1_reply_{SenatorName}.json: stores reply data for each senator. We collected data for all the senators. *Key*: Post-level dictionaries for each post within the last 7 days. *Value*: For each post, post metadata (uri, createdAt, text, likeCount, repostCount, replyCount, repliesCollected) and a list of collected replies. Each reply contains replier information (handle, displayName, timestamp, text, likeCount).
- PartII1_reply_allSenators.json: aggregates the reply data for all senators. *Key*: Each senator's metadata. *Value*: A list of posts collected within the past 7 days and their corresponding replies (up to 200 replies per post due to API limits).

II.2 Gender Inference [GENDER INFERENCE CODEs ARE SAVED IN bluesky_helpers.py IN THE PART2 FOLDER]

Based on the assignment guidance, we infer repliers' gender by extracting a likely first name from each replier's display name and matching it to SSA baby name data. We first cleaned the display names by keeping only alphabetic characters, spaces, apostrophes, and hyphens. Also, we removed titles such as Mr, Ms, Dr, and Sen. Then, we treated the first token as the replier's first name. If the extracted likely first name is too short, not alphabetic, or not found in the SSA baby name data, we classified the gender as unknown. For names in the SSA baby name data, we looked up the name's historical gender distribution by aggregating female and male name counts across all years. Lastly, we classified the replier as female if > 60% of registrations are female, male if > 60% are male, otherwise unknown. We used a 60% threshold rather than a stricter threshold to classify more repliers' gender.



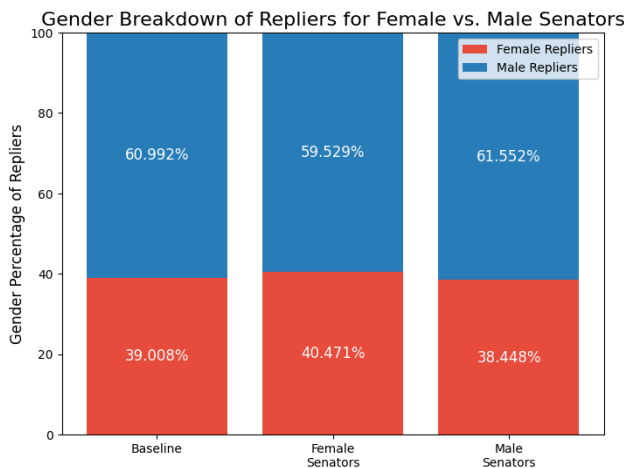
Category	Count
Total Replies Collected	13,056
Classifiable Repliers	4,599 (35.225%)
- Female	1,794 (13.741%)
- Male	2,805 (21.484%)
Unclassifiable Repliers	8,457 (64.775%)
- Non-empty Name	5,167 (39.576%)
- Empty Name	3,290 (25.199%)

Using this approach, we were able to infer the repliers' gender of approximately 35% of all replies. Out of 13,056 total replies, 4,599 replies were classified as either female or male. The 65% (8,457 replies) of replies could not be classified. Among these unclassifiable cases, 3,290 had empty display names, while 5,167 had non-empty display names but could not be matched to SSA name data or did not meet the classification threshold.

This approach has clear limitations given that about 65% of replies' gender was unclassifiable. The most significant limitation is using the displayName rather than real names. Since displayName is similar to a nickname, users may enter non-name phrases or leave it blank. For example, displayName such as "How You Like Me Now?..." or "Soulless" cannot be classified. Second, names that do not appear in the SSA dataset such as international names cannot be classified using this method. Third, the approach has a misclassification problem. Non-binary repliers may be misclassified as either female or male. Additionally, many names are used by both males and females, and using only historical probabilities may lead to misclassification. Also, some gender neutral names are classified as unknown because they do not exceed the threshold. Moreover, gender distribution of names varies across generations, but this approach does not account for this variation.

II.3 Homophily Measurement

To measure homophily, we first calculated the baseline gender distribution of classifiable repliers across all senators (p_{female} , p_{male}). Then, we calculated the fraction of same-gender repliers for female and male senators (obs_F , obs_M). Lastly, we computed the homophily coefficients as the difference between these fractions and the baseline.



Category	Female (%)	Male (%)
Baseline	39.008	60.992
Replies to Female Sen.	40.471	59.529
Replies to Male Sen.	38.448	61.552
Homophily Coef.	+0.015	+0.006

There is little evidence of gender homophily. The homophily coefficients are +0.015 for female senators and +0.006 for male senators. Although homophily coefficients are positive, they are very close to 0. This indicates extremely mild homophily (i.e., same-gender repliers are over-represented relative to baseline) or almost no preference.

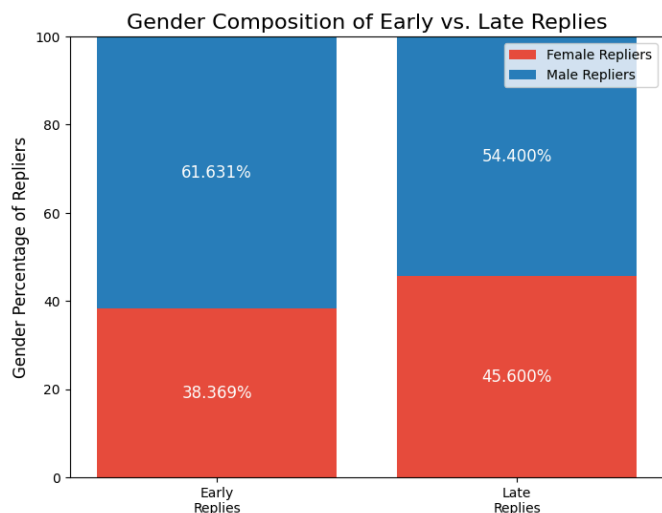
Chi-square Statistics	df	p-value	Conclusion
1.501	1	0.220	Fail to reject H0

To assess whether these differences are statistically significant, we conducted a chi-square test of independence using a contingency table of replier gender by senator gender. We chose the chi-square test because it is appropriate for testing independence between two categorical variables (senator gender and replier gender). Also, we compare female and male senators directly because comparing each group to the baseline (which combines both groups) would reuse the same data and violate independence. The null hypothesis is that replier gender is independent of senator gender (i.e., distribution of replier gender is the same for female and male senators). The test results indicate the chi-square statistic of 1.501 (df = 1,

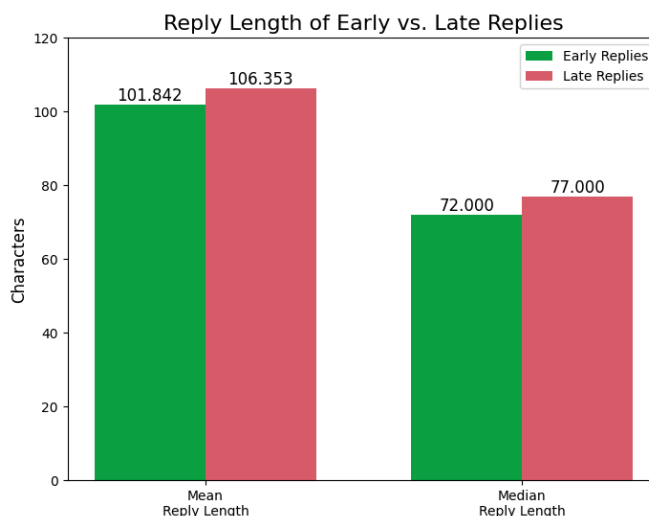
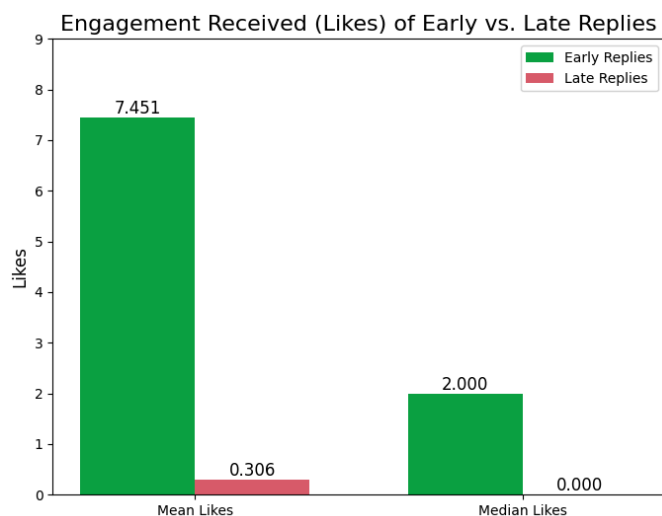
p-value = 0.220). Since the p-value is bigger than 0.05, we fail to reject the null hypothesis. It means that there is no statistically significant evidence of gender homophily, implying that the small positive homophily coefficients could plausibly arise by chance. However, we should interpret the result cautiously due to the limitations of the gender inference approach.

II.4 Reply Timing Analysis

We filtered posts with 50–200 total replies, split replies into early (first 25%) and late (last 25%) replies, and compared gender composition, reply length, and likes between early and late replies to check potential bias due to the API reply limit.



The gender composition of early and late replies shows a difference. Among early replies, the female proportion was approximately 38%, whereas among late replies it was about 46%. To assess whether this difference is statistically significant, we conducted a chi-square test using a contingency table of replier gender by reply timing. The null hypothesis is that replier gender is independent of reply timing. The test results indicate the chi-square statistic of 3.951 (df = 1, p-value = 0.047). Since the p-value is below 0.05, we reject the null hypothesis, meaning that gender composition differs significantly between early and late replies.



Reply length and engagement received (likes) also show differences between early and late replies. The average length of early replies was 102 characters (median = 72), while late replies had an average length of 106 characters (median = 77). Early and late replies showed a very mild difference in reply length. Engagement received (likes) showed a clear difference. Early replies received an average of 7.451 likes (median = 2), whereas late replies received only an average 0.306 likes (median = 0). This difference may be due to early replies being shown higher in the thread.

Senator	Gender	Posts Length Mean	Total Posts	High Engagement Posts	Percent
Chris Murphy	M	231.600	10	4	40.000
Chuck Schumer	M	211.455	22	3	13.636
Elissa Slotkin	F	206.478	46	2	4.348

The table shows senators with more than one high-engagement post (200+ replies). Among the 42 senators, 8 had at least one high-engagement post in the past seven days (Chris Murphy, Chuck Schumer, Elissa Slotkin, Mark Warner, Tina Smith, Amy Klobuchar, Ed Markey, and Adam Schiff). There were 14 high-engagement posts in total during this period. There was no clear pattern implying that these senators share a specific characteristic, such as gender, posting amount, or longer post length. However, it is interesting that only a small subset of senators have high-engagement posts. In particular, Chris Murphy had 4 high-engagement posts out of 10 total posts.

The timing analysis shows clear differences between early and late replies, indicating that bias due to the API reply limit should be acknowledged. Also, the bias is concentrated to a few senators who receive high-engagement posts, meaning that the bias affects some senators more than others. In addition, the overall reply collection coverage was 61.6%, meaning that 38.4% of replies (8,140 / 21,196) were not captured. Thus, it is necessary to include this caveat in the previous analysis. However, this is not a major caveat for our overall reply homophily analysis. Out of 645 total posts, only 14 were high-engagement posts, and the analysis was conducted at the aggregate level across senators rather than comparing individual senators. Nevertheless, as shown in the timing analysis, early and late replies differ in gender composition and engagement patterns, meaning that the limitations of the collected data may slightly distort the results. Thus, this caveat should be acknowledged.

II.5 Reflection Questions

1. The main limitation is that we rely on display names rather than real names, and many users leave them blank or non-name phrases. Also, international names not included in the SSA data cannot be classified. This approach also involves some degree of misclassification, especially for non-binary gender users or gender-neutral names. Additionally, it does not account for variation in the gender distribution of names across generations.
2. We would like to examine geographic homophily. It is reasonable to expect that users from the same state may be more likely to reply to their own state's senator. Especially in the U.S., laws and cultures vary by state, so people may feel stronger connections to their state's senators. Thus, it would be meaningful to examine geographic homophily. As a baseline, we can calculate the overall fraction of repliers from a specific state. Then, for each senator, we can calculate the fraction of repliers who are in the same state. Subtracting the baseline from this value can be used as the geographic homophily coefficient. Positive value indicates geographic homophily. If state-level classification is too narrow, the analysis could instead be conducted at the regional level (e.g., South, West). Although it may not be possible to obtain geographic information for users, geographic data could be inferred from the profile bio or user's posts.
3. For a senator whose posts routinely receive 1,000+ replies, we are capturing at most about 20% of total replies. This means that most of the replies are not captured. Based on our analysis, early and late replies differ in gender composition and engagement. Late replies show a higher proportion of female repliers and receive fewer likes. Also, reply length shows a mild difference. Thus, the missing replies may show systematic differences in gender composition (more female) and engagement (fewer likes). If some senators routinely receive 1,000+ replies, the systematic differences between early and late replies would reduce our confidence in the homophily estimates. However, in our analysis, only 14 out of 645 posts were high-engagement posts. Thus, the API reply limit may result in bias and should be recognized, but it does not severely undermine confidence in the homophily estimates.

Part III: Platform biases

1. The absence of Republican senators limits the external validity of our findings. It shows a significant selection bias towards democrats who use Bluesky, meaning our analysis reflects the behavior of the Democratic caucus rather than the broader U.S. Senate or general political figures. Result in the failure of making claims about cross-party polarization. Instead, we could focus on intra-party dynamics, treating the network as a single-party ecosystem where we can observe internal factions (e.g., Progressives vs. Moderates) and information flow without the influence of inter-party competition.
2. Because Bluesky has no Republican senators, we would consider using data from X (Twitter) or Instagram, where both Democratic and Republican senators exist and similar engagement functions (replies, likes, reposts) exist. We would then like to examine partisan homophily using a similar approach to the gender homophily analysis. First, as a baseline, we can calculate the overall fraction of engagements (replies, likes, reposts) from a specific party (D/R/I). Then, for each senator, we can calculate the fraction of engagements from users affiliated with the same party. Subtracting the baseline from this value can be used as the partisan homophily coefficient. In addition, we would also like to try a network-based assortativity analysis by labeling nodes with parties, building engagement networks, constructing a party mixing matrix (count edges), and calculating the assortativity coefficient to try alternative approaches and statistically assess partisan homophily.
3. The partisan composition of Bluesky's overall user base may cause bias, since the results can reflect the behavior of Democratic users rather than the general population. If people who share the same partisan identity tend to interact more with each other regardless of gender, the results may not measure a pure "gender" homophily pattern. In addition, because the baseline is also calculated using replies to Democratic senators' posts, the gender effect may already be mixed with the party effect. Since this baseline is derived from the same political identity group, subtracting it does not resolve the representativeness problem and could underestimate the true gender homophily. Thus, the results should be interpreted cautiously.